

TrES-3b

R-1155

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220419 · created 2026-07-31T01:38:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459688.8713 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.057 +/- 0.053
Transit depth (Rp/Rs)^2	0.33 +/- 0.6 [%]
Orbital Inclination (inc)	80.88 +/- 2.82
Airmass coefficient 1 (a1)	1.077 +/- 0.048
Airmass coefficient 2 (a2)	-0.066 +/- 0.074
Scatter in the residuals of the lightcurve fit is	4.43 %
Best Comparison Star	#3 - [477, 316]
Optimal Method	PSF photometry
Transit Duration (day)	0.027 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.057 ± 0.053 vs 0.16309 (deviation 2.0σ)
Duration fit vs published	0.027 d vs 0.0567 d (deviation 1.14σ)
Mid-time O-C	-4.6 min
Depth SNR	1.1
Residual scatter	4.43 %

Light curve

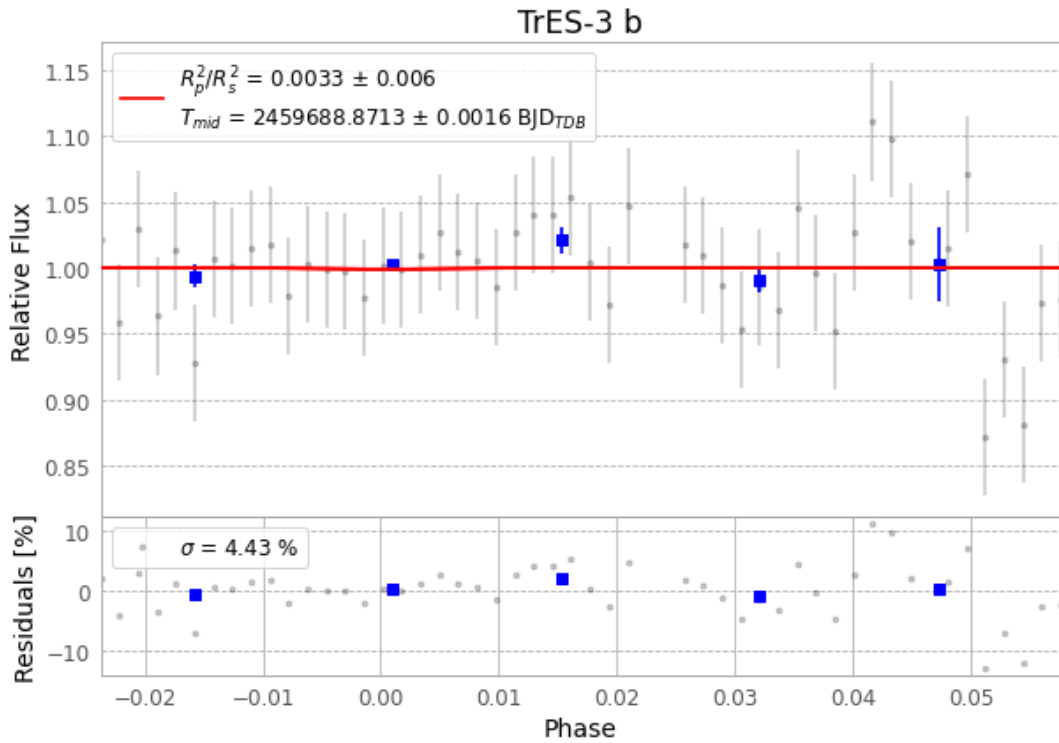


Figure 1 — FinalLightCurve_TrES-3 b_2022-04-19.png (SHA-256 6924d44e7eca7c9c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [179, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cd5ebd96064cb67d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

52 FITS frames (34 MB), TRES-3220419080613.FITS → TRES-3220419103914.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220419.log (d283a9ebfaa1...), FinalLightCurve_TrES-3 b_2022-04-19.png (6924d44e7eca...), FinalLightCurve_TrES-3 b_2022-04-19.csv (1fc0c4ddce8f...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 01:38Z.